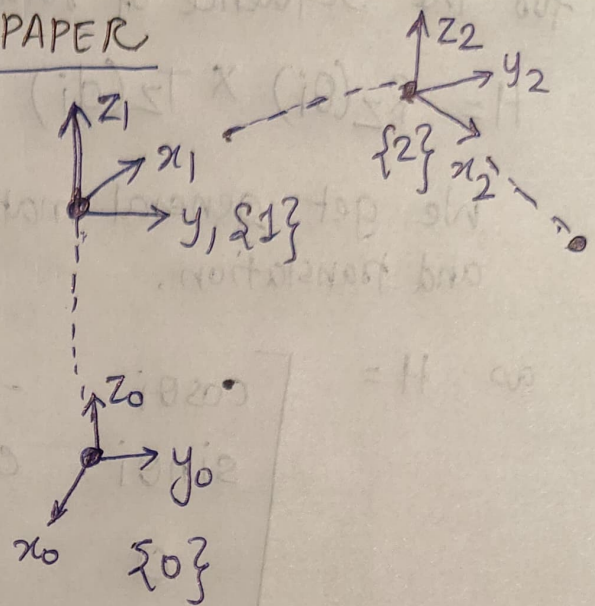
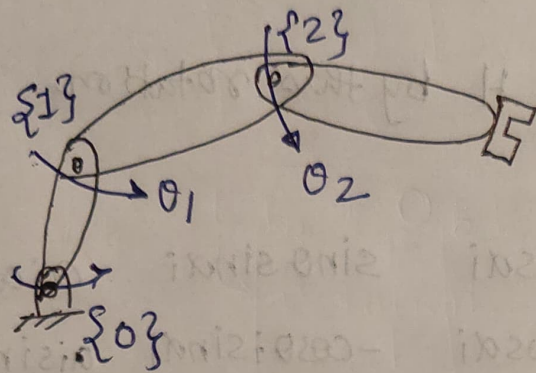


2-DOF manipulator {D-H convention FK}

TASK 1 PEN & PAPER



$\{0\} \rightarrow$ fixed frame

$\{1\} \rightarrow$ Revolute joint 1 with θ_1 frame

$\{2\} \rightarrow$ Revolute joint 2 with θ_2 frame

D-H Table by 7 steps:

	θ	α_i	a_i	d_i
link 1	θ_1	90°	a_1	0
link 2	θ_2	0°	a_2	0

$\rightarrow$ Considering Revolute rotation only no prismatic and angle of twist for link 1 as 90° & 0° for link 2

let joint variables be θ_1, θ_2

link lengths a_1, a_2

No, prismatic so $d_1, d_2 = 0$.

for the sequence of rotation

$$H = R_z(\theta_i) \times T_z(d_i) \times R_x(\alpha_i) \times T_x(a_i)$$

We get general matrix as H by this rotation and translation.

$$\text{as } H = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \cos\alpha_i & \sin\theta_i \sin\alpha_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\theta_i \cos\alpha_i & -\cos\theta_i \sin\alpha_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

General H matrix.

Now for our 2-D manipulator

Step: 1

Transformation from $\{0\}$ to $\{1\}$

$$\therefore T_1^0 = \begin{bmatrix} \cos\theta_1 & 0 & \sin\theta_1 & a_1 \cos\theta_1 \\ \sin\theta_1 & 0 & -\cos\theta_1 & a_1 \sin\theta_1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

at $\alpha = 90^\circ$
 $\theta = \theta_1$
 in H

Step: 2

Transformation from $\{1\}$ to $\{2\}$

$$T_2^1 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & a_2 \cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & 0 & a_2 \sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

at $\alpha = 0^\circ$
 $\theta = \theta_2$ in H

Step 3

$$T_2^0 = T_1^0 \times T_2^1$$

$$= \begin{bmatrix} c_1 & 0 & s_1 & a_1 c_1 \\ s_1 & 0 & -c_1 & a_1 s_1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_a^0 = \begin{bmatrix} c_1 c_2 & -c_1 s_2 & s_1 & c_1(a_1 + a_2 c_2) \\ s_1 c_2 & -s_1 s_2 & c_1 & s_1(a_1 + a_2 c_2) \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\therefore$ By $\begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix}$

End effector position.

$$x = (a_1 + a_2 \cos \theta_2) \cos \theta_1$$

$$y = (a_1 + a_2 \cos \theta_2) \sin \theta_1$$

$$z = a_2 \sin \theta_2$$

$$\therefore p = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (a_1 + a_2 \cos \theta_2) \cos \theta_1 \\ (a_1 + a_2 \cos \theta_2) \sin \theta_1 \\ a_2 \sin \theta_2 \end{bmatrix}$$

We get The position via F.K of joint variables ^{through}